Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Langworth Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
5306 Langworth RD	Oakdale	Stanislaus	95361

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1935

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X062-X027-X003-XXXX

B. OPERATORS

Silva, Jose Manuel

Operator name: Silva, Jose Manuel

Telephone no.: (209) 869-3348 Landline, (209) 765-8008 Cellular

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 2153	Oakdale	CA	95361

C. OWNERS

Silva, Jose Manuel

Legal owner name: Silva, Jose Manuel

Telephone no.: (209) 869-3348 Landline, (209) 765-8008 Cellular

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 2153	Oakdale	CA	95361

Silva, Maria

Legal owner name: Silva, Maria

Telephone no.: (209) 869-3348 Landline, (209) 765-8008 Cellular

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 2153	Oakdale	CA	95361

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	75	0
Number under roof	470	100	30	0	0	130
Maximum number	470	100	30	0	75	130
Average number	470	100	30	0	75	130
Avg live weight (lbs)	1,250	1,300	1,200	0		

Predominant milk cow breed: Jersey-Holstein Cross

Average milk production: 73 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 14,667.68 *tons per reporting period*

Total nitrogen from manure: 191,364.74 *lbs per reporting period*

After ammonia losses (30% loss applied): 133,955.32 *lbs per reporting period*

Total phosphorus from manure: 30,714.97 *lbs per reporting period*

Total potassium from manure: 90,935.71 *lbs per reporting period*

Total salt from manure: 244,294.50 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 2,268,000 *gallons*

Total nitrogen generated: 19,536.31 *lbs*

Total phosphorus generated: 1,092.27 *lbs*

Total potassium generated: 19,056.84 *lbs*

Total salt generated: 136,564.92 *lbs*

	2,268,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	2,268,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Langworth Dairy Dom. Well	Ground water
OID Langworth Dairy Irrigation	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/12/2023	Corral solids	3,460.00 *ton*	Dry-weight	62.4		15,700.00	6,390.00	13,100.00		0.00
10/30/2023	Scraped material	1,460.00 *ton*	Dry-weight	8.8		24,100.00	11,700.00	25,600.00		0.00

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	105,029.41	47,783.84	102,258.98	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	105,029.41	47,783.84	102,258.98	0.00

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field #2	51	51	6	process wastewater	X062-X027-X003-XXXX
Totals for areas that were used for application	51	51	6		
Totals for areas that were not used for application					
Land application area totals	51	51	6		

B. CROPS AND HARVESTS

Field #2

Field name: Field #2

11/17/2022: Winter Forage

Crop: Winter Forage Acres planted: 51 Plant date: 11/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/18/2023	357.00 *ton*	Dry-weight		72.2	16,800.00	2,760.00	25,600.00		11.90
05/18/2023	357.00 *ton*	Dry-weight		73.0	16,200.00	2,690.00	25,200.00		13.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	148.80	24.00	224.00	0.00
Total actual harvest content	14.00	126.62	20.91	194.89	962.11

Field #2

05/23/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 51 Plant date: 05/23/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/24/2023	255.00 *ton*	Dry-weight		82.5	19,100.00	5,230.00	33,400.00		12.80
08/24/2023	255.00 *ton*	Dry-weight		74.4	14,200.00	37,400.00	18,400.00		9.44
10/30/2023	204.00 *ton*	Dry-weight		63.4	16,300.00	2,070.00	16,800.00		15.60
10/30/2023	204.00 *ton*	Dry-weight		61.0	14,700.00	2,060.00	15,200.00		8.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	216.00	0.00
Total actual harvest content	18.00	163.37	117.38	202.17	1,187.63

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field #2 - 11/17/2022: Winter Forage

Field name: Field #2

Crop: Winter Forage — Plant date: 11/17/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/15/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	83.29	6.70	112.98	606.97	504,000.00 *gal*
Application event totals		83.29	6.70	112.98	606.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	84.94	4.60	45.60	667.99	504,000.00 *gal*
OID Langworth Dairy Irrigation	Surface water	0.00	0.00	0.00	0.00	5,832,000.00 *gal*
Application event totals		84.94	4.60	45.60	667.99	

Field #2 - 05/23/2023: Sudangrass, silage

Field name: Field #2

Crop: Sudangrass, silage — Plant date: 05/23/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID Langworth Dairy Irrigation	Surface water	0.00	0.00	0.00	0.00	5,454,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Field #2 - 05/23/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	84.94	4.60	45.60	667.99	504,000.00 *gal*
OID Langworth Dairy Irrigation	Surface water	0.00	0.00	0.00	0.00	4,752,000.00 *gal*
Application event totals		84.94	4.60	45.60	667.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID Langworth Dairy Irrigation	Surface water	0.00	0.00	0.00	0.00	5,184,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	86.59	3.68	112.98	489.86	504,000.00 *gal*
OID Langworth Dairy Irrigation	Surface water	0.00	0.00	0.00	0.00	4,752,000.00 *gal*
Application event totals		86.59	3.68	112.98	489.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID Langworth Dairy Irrigation	Surface water	0.00	0.00	0.00	0.00	5,184,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	43.30	1.84	56.49	244.93	252,000.00 *gal*
OID Langworth Dairy Irrigation	Surface water	0.00	0.00	0.00	0.00	4,752,000.00 *gal*
Application event totals		43.30	1.84	56.49	244.93	

Field #2 - 05/23/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID Langworth Dairy Irrigation	Surface water	0.00	0.00	0.00	0.00	5,184,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

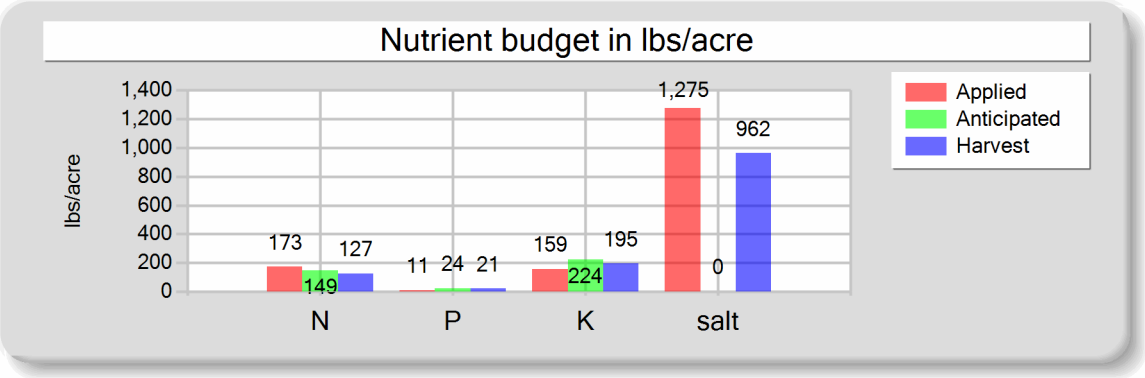
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
OID Langworth Dairy Irrigation	Surface water	0.00	0.00	0.00	0.00	4,752,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

B. NUTRIENT BUDGET

Field #2 - 11/17/2022: Winter Forage

Field name: Field #2 Crop: Winter Forage Plant date: 11/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	168.24	11.30	158.59	1,274.96
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	172.90	11.30	158.59	1,274.96
Anticipated crop nutrient removal	148.80	24.00	224.00	0.00
Actual crop nutrient removal	126.62	20.91	194.89	962.11
Nutrient balance	46.28	-9.61	-36.30	312.85
Applied to removed ratio	1.37	0.54	0.81	1.33

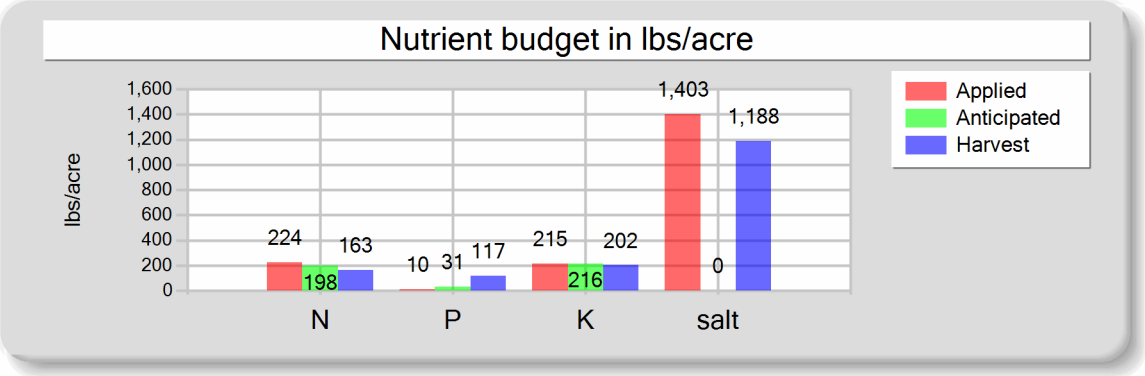
Fresh water applied	
5,832,000.00	*gallons*
214.77	*acre-inches*
4.21	*inches/acre*

Process wastewater applied	
1,008,000.00	*gallons*
37.12	*acre-inches*
0.73	*inches/acre*

Total harvests for the crop	
2	*harvests*

Field #2 - 05/23/2023: Sudangrass, silage

Field name: Field #2 Crop: Sudangrass, silage Plant date: 05/23/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	214.83	10.12	215.08	1,402.78
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	9.33	0.00	0.00	0.00
Total nutrients applied	224.16	10.12	215.08	1,402.78
Anticipated crop nutrient removal	198.00	30.60	216.00	0.00
Actual crop nutrient removal	163.37	117.38	202.17	1,187.63
Nutrient balance	60.80	-107.27	12.91	215.15
Applied to removed ratio	1.37	0.09	1.06	1.18

Fresh water applied	
40,014,000.00	*gallons*
1,473.58	*acre-inches*
28.89	*inches/acre*

Process wastewater applied	
1,260,000.00	*gallons*
46.40	*acre-inches*
0.91	*inches/acre*

Total harvests for the crop	
4	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

T3139071D

Sample and source description: T3139071D

Sample date: 05/19/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,700.00	6,390.00	13,100.00							0.00
DL	1.00	1.00	1.00							1.00

T3242048F

Sample and source description: T3242048F

Sample date: 08/30/2023 Material type: Scraped material Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 8.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,100.00	11,700.00	25,600.00							0.00
DL	1.00	1.00	1.00							1.00

B. PROCESS WASTEWATER ANALYSES

T3093011A

Sample and source description: T3093011A

Sample date: 03/31/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,010.00	619.00	0.00	0.00	81.20	1,370.00								10,300.00	7,360
DL	1.00	1.00	1.00	1.00	1.00	1.00								100.00	1

T3139071C

Sample and source description: T3139071C

Sample date: 05/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,030.00	588.00	0.00	0.00	55.80	553.00								10,200.00	8,100
DL	1.00	1.00	1.00	1.00	1.00	1.00								100.00	1

T3242048C

Sample and source description: T3242048C

Sample date: 08/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,050.00	585.00	0.00	0.00	44.60	1,370.00								10,200.00	5,940
DL	1.00	1.00	1.00	1.00	1.00	1.00								100.00	1

C. FRESH WATER ANALYSES

Langworth Dairy Dom. Well

T3242048E

Sample description: T3242048E

Sample date: 08/30/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	26.30	0.00	26.30								949.00	0
DL	1.00	0.50	1.00								100.00	1

OID Langworth Dairy Irrigation

OID Langworth Dairy Irrigation

T3242048D

Sample description: T3242048D

Sample date: 08/30/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	0
DL	0.14	0.50	0.14								100.00	1

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Field #2 - 11/17/2022: Winter Forage

T3139071A

Sample and source description: T3139071A

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,800.00	2,760.00	25,600.00		11.90
DL	1.00	1.00	1.00		1.00

T3139071B

Sample and source description: T3139071B

Sample date: 05/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,200.00	2,690.00	25,200.00		13.20
DL	1.00	1.00	1.00		1.00

Field #2 - 05/23/2023: Sudangrass, silage

T3242048A

Sample and source description: T3242048A

Sample date: 08/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 82.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,100.00	5,230.00	33,400.00		12.80
DL	1.00	1.00	1.00		1.00

T3242048B

Sample and source description: T3242048B

Sample date: 08/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,200.00	37,400.00	18,400.00		9.44
DL	1.00	1.00	1.00		1.00

T3303034B

Sample and source description: T3303034B

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,300.00	2,070.00	16,800.00		15.60
DL	1.00	1.00	1.00		1.00

Field #2 - 05/23/2023: Sudangrass, silage

T3303034C

Sample and source description: T3303034C

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,700.00	2,060.00	15,200.00		8.50
DL	1.00	1.00	1.00		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

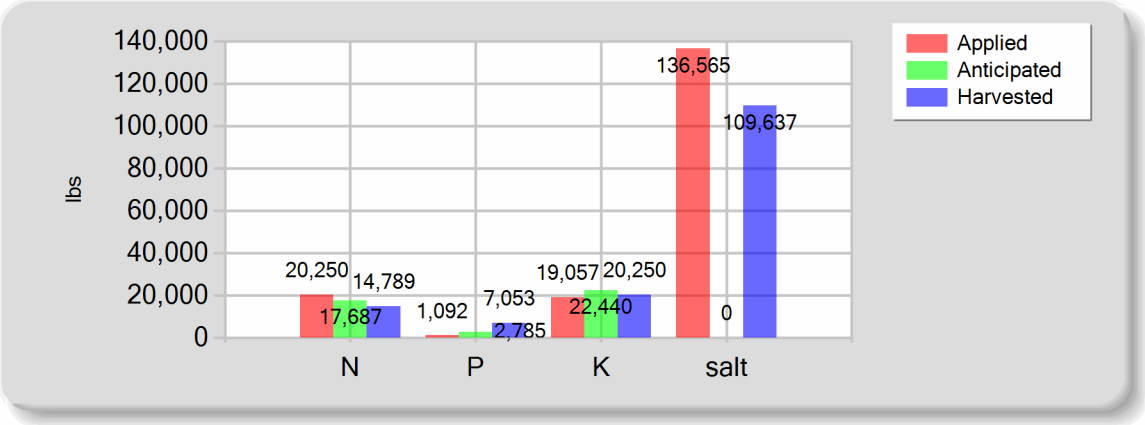
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

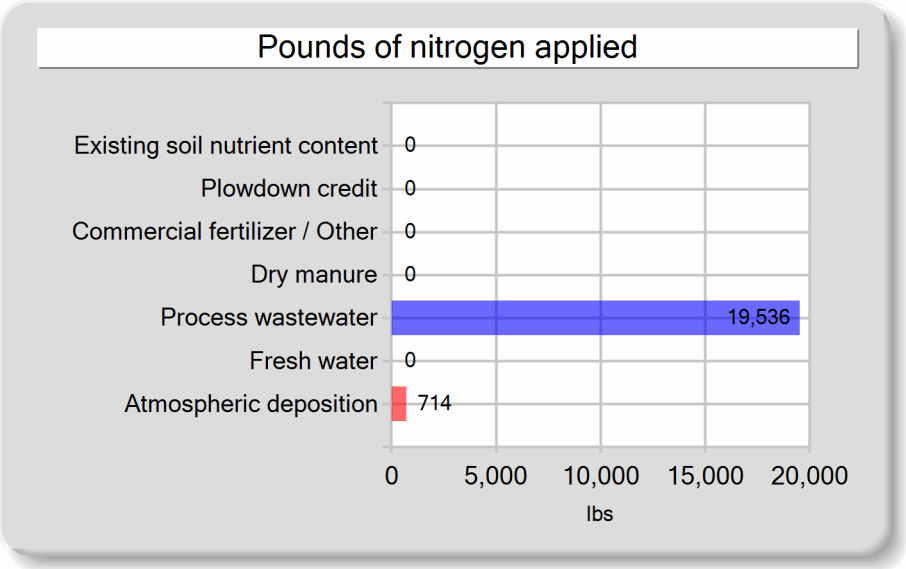
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

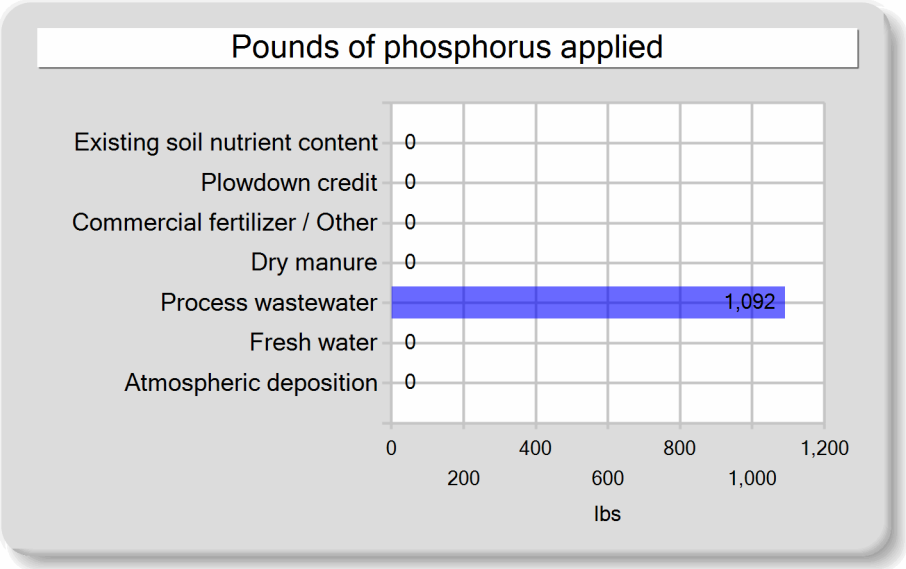
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	19,536.31	1,092.27	19,056.84	136,564.92
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	714.00	0.00	0.00	0.00
Total nutrients applied	20,250.31	1,092.27	19,056.84	136,564.92
Anticipated crop nutrient removal	17,686.80	2,784.60	22,440.00	0.00
Actual crop nutrient removal	14,789.44	7,053.03	20,250.04	109,636.74
Nutrient balance	5,460.87	-5,960.77	-1,193.20	26,928.18
Applied to removed ratio	1.37	0.15	0.94	1.25

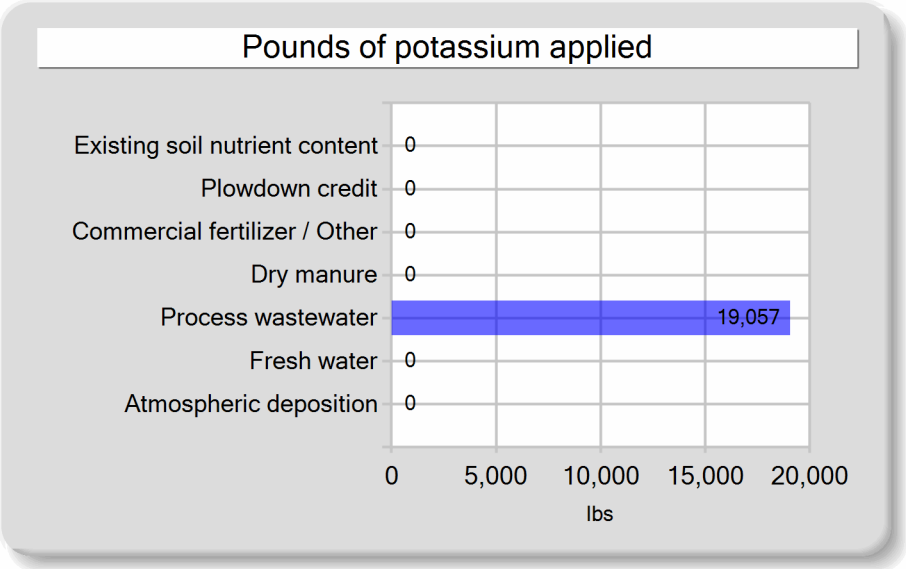
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

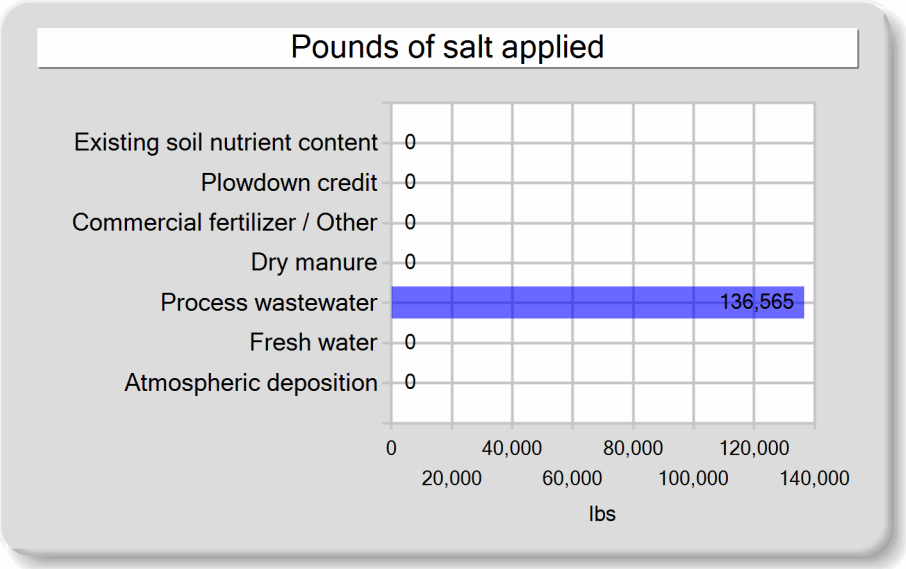


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Maria Silva	Jose Silva
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Maria Silva	Jose Manuel Silva
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6-12-24	6-12-24
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: ____Jose Silva_____

Name of Dairy Facility: __Langworth Dairy_____

Facility Address: _5306 Langworth Rd._____Oakdale_____95361__
Number and Street City Zip Code

Contact Person Name and Phone Number: __Maria Silva_____209-735-1858
Name Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: _Caton Ag, Inc._____

Address of Hauling Company /Person: _14303 Campbell Rd._____Escalon_____95361_
Number and Street City Zip Code

Contact Person: __Matt Caton_____209-495-7769_____
Name Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____(please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

_____SAME_____
Name Number and Street City Zip Code Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

_____SAME_____
Number and Street City Zip Code Assessor's Parcel Number

Dates Hauled: ____May 1-12, 2023_____

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount: N 1.57%
P 0.639% K 1.31%

Manure: ___3460_____Tons or Cubic Yards (indicate which units used)
Manure Solids Content (if amount reported in tons): ___62.4%_____
Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: ____Estimation given by Hauler

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Marin Silva Date: 8/16/23

Hauler's Signature: _____ Date: 8/16/23

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Langworth Dairy
Mailing address: 5306 Langworth Rd.
(not facility address) Oakdale, CA 95361
Phone#: 209-735-1858

Authorized Copy Released To: Jackie

Sampled By:
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	8-30-23	9:30 AM	Langworth Dairy	Sudan				M.S ~~X~~		X	
2	8-30-23	9:45 AM	Langworth Dairy	Sudan				M.S ~~X~~		X	
3	8-30-23	10:00 AM	Langworth Dairy	Lagoon		22.0'		X			
4	8-30-23	10:15 AM	OTD Langworth Dairy	Irrigation		21.0'	X				
5	8-30-23	10:30 AM	Langworth Dairy	Dom well		21.2'	X				
6	8-30-23	10:45 AM	Langworth Dairy	Export manure		~~22.0'~~			X		
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged in By:	LABORATORY NUMBER
1st	Marcos Silva	Langworth Dairy				
2nd		DENELE ANALYTICAL, INC.	8-30-23 1:45			T3242048

Required to be submitted to any agency? (YES / NO)

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
LANGWORTH DAIRY
PO BOX 2153
Oakdale, CA 95361

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 8/30/23 1:40 PM
SUBMITTED BY: Maria
ANALYZED DATE: 8/30/23

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 21.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
OID Langworth Dairy Irrigation	T3242048D	Electrical Conductivity (EC)	EPA 120.1 and	< 0.100	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	40.4	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.4	94.0 (T3241014A)	92.2 (T3241014A)	1.93
Electrical Conductivity (EC)	96.3			3.35
Nitrate Nitrogen (NO3-N)	98.5	99.5 (T3241011A)	95.9 (T3241011A)	3.66

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

Denele Analytical, Inc.

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Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
LANGWORTH DAIRY
PO BOX 2153
Oakdale, CA 95361

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 8/30/23 1:40 PM
SUBMITTED BY: Maria
ANALYZED DATE: 8/30/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 21.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Langworth Dairy Dom. Well	T3242048E	Electrical Conductivity (EC)	EPA 120.1 and	0.949	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	26.3	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	607	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	98.4	94.0 (T3241014A)	92.2 (T3241014A)	1.93
Electrical Conductivity (EC)	96.3			3.35
Nitrate Nitrogen (NO3-N)	98.5	99.5 (T3241011A)	95.9 (T3241011A)	3.66

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